

# **ECEN 4/5632 Theory and Application of Digital Filters**

**Lecture 2, 8-25-26**

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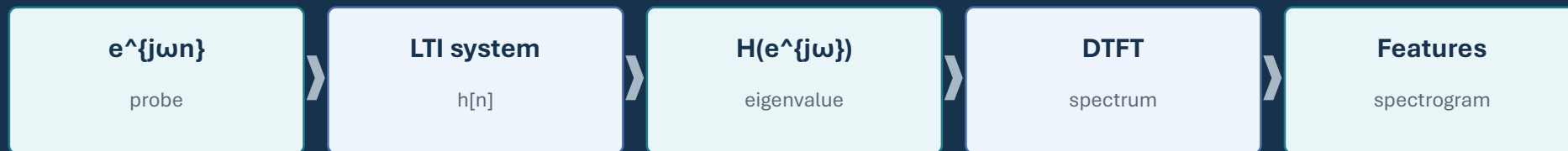


\* With generous help from Gemini Pro and ChatGPT

# Eigenfunctions and the DTFT

Week 1 • Lecture 2 • 75 minutes

Why sinusoids survive LTI systems—and why frequency-domain multiplication is so powerful



# Today's learning objectives

00-10 MIN

Move from time-domain mechanics to frequency-domain structure.

- **Prove that complex exponentials are eigenfunctions of LTI systems.**
- Define the DTFT and inverse DTFT and explain  $2\pi$  periodicity.
- Use time shift, convolution, and Parseval properties.
- Derive frequency response directly from a linear constant-coefficient difference equation.
- Interpret a one-pole filter in time and frequency domains.

## Graduate expectation

Be able to state convergence or stability assumptions when a transform/frequency response is used.

## Recap: an LTI system is convolution by $h[n]$

00-10 MIN

Now choose an input that makes convolution simplify dramatically.

$$y[n] = \sum_k h[k] x[n - k]$$

$$x[n] = e^{j\omega n}$$

$$y[n] = \sum_k h[k] e^{j\omega(n-k)}$$

$$= e^{j\omega n} \sum_k h[k] e^{-j\omega k}$$

### Notice

The input waveform  $e^{j\omega n}$  factors back out. The system can only multiply it by a complex number.

# Complex exponentials are eigenfunctions of every LTI system

00-10 MIN

The eigenvalue depends on frequency.

$$y[n] = H(e^{j\omega})e^{j\omega n}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

## What stays the same

- Frequency  $\omega$
- Exponential form
- Infinite-duration mode

## What changes

- Magnitude:  $|H(e^{j\omega})|$
- Phase:  $\angle H(e^{j\omega})$
- Possibly complete cancellation

# Frequency response is a gain-and-phase lookup table

10-35 MIN

One complex number for each normalized frequency  $\omega$ .

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$



## Audio EQ

A graphic equalizer is approximately a deliberately shaped  $|H(e^{j\omega})|$ .

# The DTFT analyzes a sequence as a continuous function of frequency

10-35 MIN

Time is discrete; frequency remains continuous.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

## Forward DTFT

- Sequence  $x[n] \rightarrow$  function  $X(e^{j\omega})$
- Weighted sum of rotating phasors
- Requires convergence in an appropriate sense

## Inverse DTFT

- Integrates one period of spectrum
- Reconstructs each integer sample
- Any interval of length  $2\pi$  works

## Why is the DTFT $2\pi$ -periodic?

10-35 MIN

Because sampled-time exponentials cannot distinguish  $\omega$  from  $\omega+2\pi k$ .

$$e^{-j(\omega+2\pi)n} = e^{-j\omega n} e^{-j2\pi n} = e^{-j\omega n}$$

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$



**same spectral content repeats every  $2\pi$**



# Normalized frequency $\leftrightarrow$ physical frequency

10-35 MIN

Week 2 will make this relationship central.

$$\omega = 2\pi \frac{f}{f_s} \iff f = \frac{\omega}{2\pi} f_s$$

$$\omega = \pi \iff f = \frac{f_s}{2}$$

## Example

At  $f_s=48$  kHz:  $\omega=\pi/2 \leftrightarrow 12$  kHz, and  $\omega=\pi \leftrightarrow 24$  kHz.

- The interval  $-\pi \leq \omega < \pi$  represents one unique frequency period.
- Frequencies outside that interval wrap into it modulo  $2\pi$ .

## Two DTFT pairs worth knowing immediately

10–35 MIN

Use them as anchors—not as a substitute for reasoning.

$$\delta[n] \longleftrightarrow 1$$

$$a^n u[n] \longleftrightarrow \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1$$

$$\delta[n - n_0] \longleftrightarrow e^{-j\omega n_0}$$

### Why the condition?

For  $|a| < 1$ , the geometric series is absolutely summable;  $|a| \geq 1$  does not converge on the unit circle in the ordinary sense.

# Time shifting becomes linear phase

35-55 MIN

Delays are easy to recognize in the frequency domain.

$$x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$$

## Magnitude

- $|e^{-j\omega n_0}| = 1$
- A pure delay does not change spectral magnitude

## Phase

- Adds  $-\omega n_0$  radians
- Linear phase slope encodes delay

### Audio connection

An ideal sample delay preserves the spectrum's magnitude exactly; it only rotates phase versus frequency.

# Convolution in time $\leftrightarrow$ multiplication in frequency

35-55 MIN

This is one of the most important DSP simplifications.

$$y[n] = x[n] * h[n]$$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$



## Computational echo

FFT-based convolution exploits the same multiplication idea numerically for long signals.

# Parseval: energy is preserved across representations

35–55 MIN

The spectrum redistributes energy; it does not create it.

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

## Time-domain view

- Energy = sum of squared samples
- Natural for waveforms

## Frequency-domain view

- Energy density across  $\omega$
- Natural for bands and features

### ML bridge

Many audio features—band energies, log spectra, mel features—summarize how signal energy is distributed across frequency.

## Difference equations give $H(e^{j\omega})$ directly

35-55 MIN

Use the eigenfunction or DTFT property instead of solving every input separately.

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m] H(e^{j\omega}) = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

- Substitute  $x[n]=e^{j\omega n}$ ,  $y[n]=H(e^{j\omega})e^{j\omega n}$ .
- Factor out  $e^{j\omega n}$ .
- Solve the remaining algebraic equation for  $H(e^{j\omega})$ .

## One-pole example: $y[n]=0.5y[n-1]+x[n]$

A simple recursive smoother with a strong low-frequency preference.

$$y[n] - 0.5y[n-1] = x[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$

$$h[n] = (0.5)^n u[n]$$

### At $\omega=0$

- $H=1/(1-0.5)=2$
- DC is amplified

### At $\omega=\pi$

- $H=1/(1+0.5)=2/3$
- Nyquist-frequency alternation is attenuated

# Think–Pair–Share: sketch $|H(e^{j\omega})|$ before computing

55–75 MIN

Reason from a few strategically chosen frequencies.

**Given  $y[n] = 0.5y[n-1] + x[n]$**

- 1 minute: derive  $H(e^{j\omega})$ .
- 2 minutes: evaluate  $|H|$  at  $\omega=0, \pi/2, \pi$ .
- 2 minutes: sketch a smooth  $2\pi$ -periodic magnitude curve.
- Pair discussion: what audio behavior would you expect?

## Prompt

What feature of the recursion causes “memory” of prior samples?

## Graduate extension

Relate the coefficient 0.5 to a pole at  $z=0.5$  and explain why moving it toward 1 increases low-frequency selectivity.

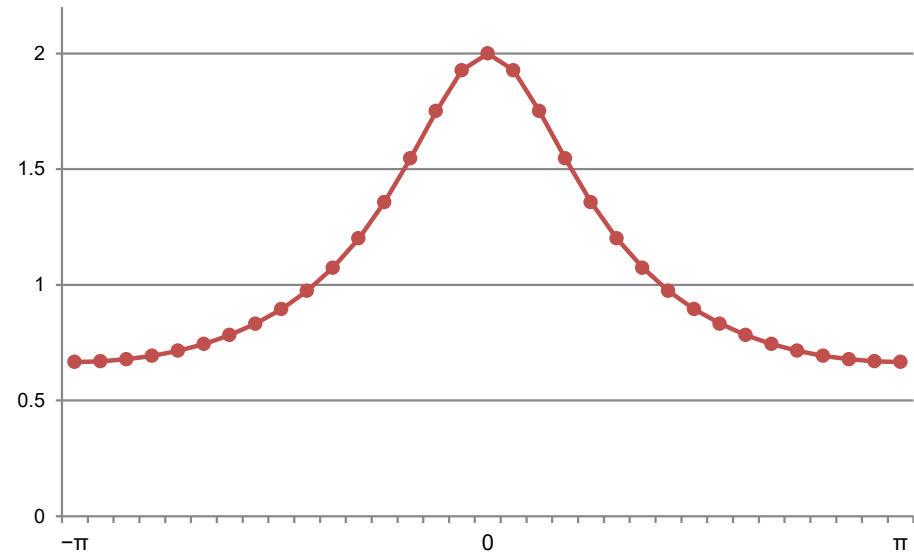


# Think-Pair-Share solution

55-75 MIN

The denominator is smallest near  $\omega=0$ , so the response peaks at low frequency.

$$|H(e^{j\omega})| = \frac{1}{\sqrt{1.25 - \cos \omega}}$$



## Three checkpoints

$\omega=0$ : 2 •  $\omega=\pi/2$ : 0.894 •  $\omega=\pi$ : 0.667

## Interpretation

Smooth low-pass tendency • recursive memory • stable because  $|0.5| < 1$

# Python demo: freqz and filtering

55–75 MIN

The numerical tools should confirm—not replace—the analysis.

## SciPy frequency response

```
import numpy as np
from scipy import signal

b = [1.0]
a = [1.0, -0.5]
w, H = signal.freqz(b, a, worN=1024)
mag = np.abs(H)
```

## Filtering

```
x = np.random.randn(48000)
y = signal.lfilter(b, a, x)
# Compare spectra / listen if desired
```

## Notebook task

Change the recursion coefficient from 0.5  $\rightarrow$  0.9  $\rightarrow$  0.99. Predict the shape first; then verify with freqz.

# DTFT vs DFT vs FFT

55–75 MIN

Three related terms that students often collapse into one.

## DTFT

- Input: discrete-time sequence
- Frequency: continuous  $\omega$
- $2\pi$ -periodic spectrum
- Mathematical transform

## DFT / FFT

- Input: finite N samples
- Frequency: N discrete bins
- DFT is the transform
- FFT is an algorithm for the DFT

### Practical bridge

In code, we sample the DTFT-like spectrum on a finite grid using the DFT/FFT. The grid and window matter.

# ML bridge: spectrograms are structured frequency-domain features

The DTFT is the conceptual ancestor of the STFT.



- DSP explains what the axes and magnitudes mean.
- Windowing and sampling choices determine resolution and leakage.
- ML operates on features whose structure comes directly from signal-processing choices.

## Concept check

CHECK

Reason from the denominator, not from memorized filter labels.

For  $H(e^{j\omega}) = 1/(1 - 0.9e^{-j\omega})$ , where is  $|H|$  largest?

A. Near  $\omega=0$

B. Near  $\omega=\pi/2$

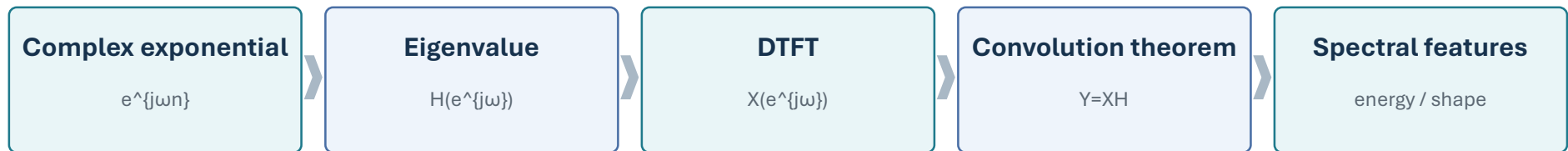
C. Near  $\omega=\pi$

D. It is constant for all  $\omega$

# Lecture 2 synthesis

WRAP-UP

The frequency domain turns LTI systems into multiplication.



- Be able to derive  $H(e^{j\omega})$  from  $h[n]$  or a difference equation.
- Be able to explain  $2\pi$  periodicity and the relation  $\omega=2\pi f/f_s$ .
- Be able to reason qualitatively about a response before plotting it.

## Week 2

Sampling will explain which analog frequencies map to the same discrete-time frequency—and why aliasing occurs.